

Where to find up to date info on how much energy AI data centers use?

Posted by u/beuef • 0 points • 15 comments

People here used to say it was negligible how much energy AI uses but I'm wondering if that has gone up in the past year

Comments

u/oldjar7 • 1 points • 2024-09-30 14:05 UTC

It's not negligible but it's not that high either.

u/Hadookan_01 • 6 points • 2024-09-29 19:46 UTC

I just wrote about it in our company blog. It's concise and you'll get a good feel about the topic: <https://blog.datafund.net/on-the-quest-to-sate-ais-insatiable-hunger-for-energy-0dee8b8349f2>

u/beuef • 1 points • 2024-09-30 00:13 UTC

I'm curious what your personal take is on this, if you think it's worth it to develop AI despite the environmental impacts

u/Hadookan_01 • 1 points • 2024-10-21 18:53 UTC

I think it is beginning to border on the ludicrous. I mean, in my home country we're in the middle of a national debate about building a 1 or 1,6 GW nuclear reactor. It's a project for the next 80 to 100 years that would supply about 40 % of the country's electricity. And here we have Google, OAI, Oracle, Microsoft etc. casually discussing and investing insane amounts of money and installing 1GW reactors just to run a datacentre? I get it, I'm all pro AI and a big fan, but it wouldn't hurt to just slow down a bit. Give more time to play through the possible shitty scenarios, invest those tens of billions of €€€ into making lives better for everyone. As some guy recently pointed out, we're actually living in postscarcity since the 19th century (I would say since 1950s). With a more equal distribution, a dream of an AI utopia would actually be quite moot. To actually answer your question, no, I don't think it's worth developing AI despite the environmental impact. Too many unknowns. Tech bros and all the fan boys talk like utopia is just 1 TW datacentre away. It was the same with the Web. I remember how it was supposed to bring the whole world together in an almost paradisaical way. It has done a lot, but far from the prophecies. Just look at all the shit online. The same with crypto, and now it feels more like a giant casino. So, no, I don't think it's worth it, because we're more likely to end up in an Elysium style world than in an utopia.

u/beuef • 1 points • 2024-09-29 20:48 UTC

Thank you very much that's interesting

u/llamatastic • 3 points • 2024-09-29 13:19 UTC

One rough method would be to count the GPUs. Nvidia was [projected](<https://www.tomshardware.com/news/nvidia-to-reportedly-triple-output-of-compute-gpus-in-2024-up-to-2-million-h100s>) to sell around 2-2.5 million H100s by the end of the year. Thermal design power is 700W, but power draw would be about 2x higher if you include overhead. So that adds up to roughly 3 GW.

u/Duarteeeee • 3 points • 2024-09-29 08:52 UTC

Perplexity AI :

The energy consumption of AI data centers has indeed increased significantly in recent years, and this trend has accelerated in the past year. Here are some key points about the current state of AI energy usage and where to find up-to-date information:

Recent Estimates and Projections

The International Energy Agency (IEA) forecasts that global data center electricity demand will more than double from 2022 to 2026, with AI playing a major role in this surge[1]. Currently, data centers account for over 1% of global electricity use, but this percentage is expected to rise.

Goldman Sachs Research estimates that:

- Data center power demand will grow 160% by 2030
- AI will represent about 19% of data center power demand by 2028
- Overall data center power consumption from AI will increase by about 200 terawatt-hours per year between 2023 and 2030[2]

Comparison to Previous Estimates

The notion that AI energy use was negligible is no longer accurate. Some key points:

- A single ChatGPT query requires 2.9 watt-hours of electricity, compared to 0.3 watt-hours for a Google search[2].
- AI-related tasks consume significantly more energy than standard cloud-based applications[3].
- The computational power required for AI is doubling roughly every 100 days[4].

Sources for Up-to-Date Information

To stay informed about the latest data on AI energy consumption, consider these sources:

1. **International Energy Agency (IEA)**: Regularly publishes reports on global energy trends, including data center and AI energy use.
2. **Research from financial institutions**: Goldman Sachs, Morgan Stanley, and other financial firms produce detailed analyses on the energy implications of AI.
3. **Tech company reports**: Major tech companies like Google, Microsoft, and Meta often release sustainability reports that include data on their energy consumption.
4. **Academic studies**: Universities and research institutions frequently publish studies on the environmental impact of AI.
5. **Industry publications**: Websites and magazines focused on technology and energy sectors often feature articles on this topic.
6. **Government agencies**: Entities like the U.S. Energy Information Administration provide data on energy consumption across various sectors.

Challenges in Measuring AI Energy Use

It's important to note that measuring the exact energy consumption of AI can be challenging due to:

- The rapid pace of technological change
- Varying definitions of what constitutes "AI" energy use
- Differences in reporting methods among companies
- The complex interplay between AI efficiency improvements and increased usage

As AI continues to evolve and expand, staying updated on its energy implications will be crucial for understanding its broader environmental impact.

Citations :

- [1] How AI Is Fueling a Boom in Data Centers and Energy Demand | TIME <https://time.com/6987773/ai-data-centers-energy-usage-climate-change/>
- [2] AI is poised to drive 160% increase in data center power demand <https://www.goldmansachs.com/insights/articles/AI-poised-to-drive-160-increase-in-power-demand>
- [3] Data center emissions probably 662% higher than big tech claims ... <https://www.theguardian.com/technology/2024/sep/15/data-center-gas-emissions-tech>
- [4] Powering Progress: The Impact of AI on Global Electricity Demand <https://leowealth.com/insights/powering-progress-the-impact-of-ai-on-global-electricity-demand/>
- [5] AI Power Consumption: Rapidly Becoming Mission-Critical - Forbes <https://www.forbes.com/sites/bethkindig/2024/06/20/ai-power-consumption-rapidly-becoming-mission-critical/>
- [6] The Artificial Intelligence energy paradox to data centers - SWEP <https://www.swep.net/company/news-and-media/news/2024/august/artificial-intelligence-energy-paradox/>
- [7] US Data Center Power Use Will Double by 2030 Because of AI <https://carboncredits.com/us-data-centers-power-requirement-will-double-by-2030/>
- [8] How data centers and the energy sector can sate AI's hunger for power <https://www.mckinsey.com/industries/private-capital/our-insights/how-data-centers-and-the-energy-sector-can-sate-ais-hunger-for-power>
- [9] Google's Report Shifts Focus onto Data Centre Emissions <https://datacentremagazine.com/technology-and-ai/googles-report-shifts-focus-onto-data-centre-emissions>
- [10] A.I.'s Insatiable Appetite for Energy - The New York Times <https://www.nytimes.com/2024/07/11/climate/artificial-intelligence-energy-usage.html>

u/Economy-Fee5830 • 8 points • 2024-09-29 09:19 UTC

You know, the claim that chatgtp uses 10x more energy than a google search comes from a source which is completely offline.

<https://www.epri.com/research/products/3002028905>

Just because you have a statement on a website does not make it properly sourced or true.

u/Duarteeeee • 1 points • 2024-09-29 09:22 UTC

It's here : <https://www.goldmansachs.com/insights/articles/AI-poised-to-drive-160-increase-in-power-demand>

u/Economy-Fee5830 • 4 points • 2024-09-29 09:23 UTC

Yes, but where did they get it ?

From here: <https://www.epri.com/research/products/3002028905>

Like I said, primary sources.

u/koeless-dev • -1 points • 2024-09-29 12:28 UTC

I mean it's nice to have primary sources sure, however Goldman Sachs is *chef's kiss* quality, so that's great already.

u/Economy-Fee5830 • 2 points • 2024-09-29 12:34 UTC

Sorry, but no - they did not even link back to the source of their claim, I had to google it. Secondly we don't know how the EPRI came to their figure, and that is pretty important.

u/koeless-dev • -1 points • 2024-09-29 14:33 UTC

Site's working again, can access their data directly. Got me interested even though I generally recommend against digging through this as non-experts. Leave it to the experts, but still it was a fun read and looks good.

u/akko_7 • 1 points • 2024-09-30 09:27 UTC

Lmao

u/Economy-Fee5830 • 2 points • 2024-09-29 14:39 UTC

So this is the primary source of the number:

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Alphabet's chairman indicated in February 2023 that interacting with an LLM could "likely cost 10 times more than a standard keyword search.⁶" As a standard Google search reportedly uses 0.3 Wh of electricity,⁹ this suggests an electricity consumption of approximately 3 Wh per LLM interaction. This figure aligns with SemiAnalysis' assessment of ChatGPT's operating costs in early 2023, which estimated that

ChatGPT responds to 195 million requests per day, requiring an estimated average electricity consumption of 564 MWh per day, or, at most, 2.9 Wh per request.

<https://asociace.ai/wp-content/uploads/2023/10/ai-spotreba.pdf>

So almost certainly an outdated number that was vague from the start and based on old hardware.

